

WEAVE: Wavelet-Decoupled Velocity Matching for Lightweight Style Transfer

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Abstract

For style transfer, SOTA diffusion models and training-free variants built on them exceed the VRAM limits of consumer GPUs. Meanwhile, lightweight alternatives can underperform a trivial identity mapping (copying the source), which scores high on traditional metrics but is rarely evaluated as a baseline. We trace this to frequency dominance in shared representation spaces: low-frequency structural energy overwhelms high-frequency style gradients and can push optimization toward identity-like solutions.

We argue that an effective method must outperform this identity baseline while remaining computationally affordable. To this end, we propose WEAVE, a wavelet-decoupled framework op-

erating in the VAE latent space. A single-level Haar transform splits the latent into a low-frequency structure band, where a lightweight velocity model handles source-aligned transport, and high-frequency style bands where stepwise statistical alignment injects target style. This decoupling separates content transport from style injection, letting WEAVE train in under two minutes on a single consumer GPU.

We also advocate the IDT–TGT sandwich as an evaluation guideline. IDT returns the source unchanged, while TGT returns an unrelated image from the target style; together they contextualize valid transfer. Under this guideline, WEAVE exceeds the identity reference on the reported board with 1.04M trainable parameters (plus a frozen 84M VAE).

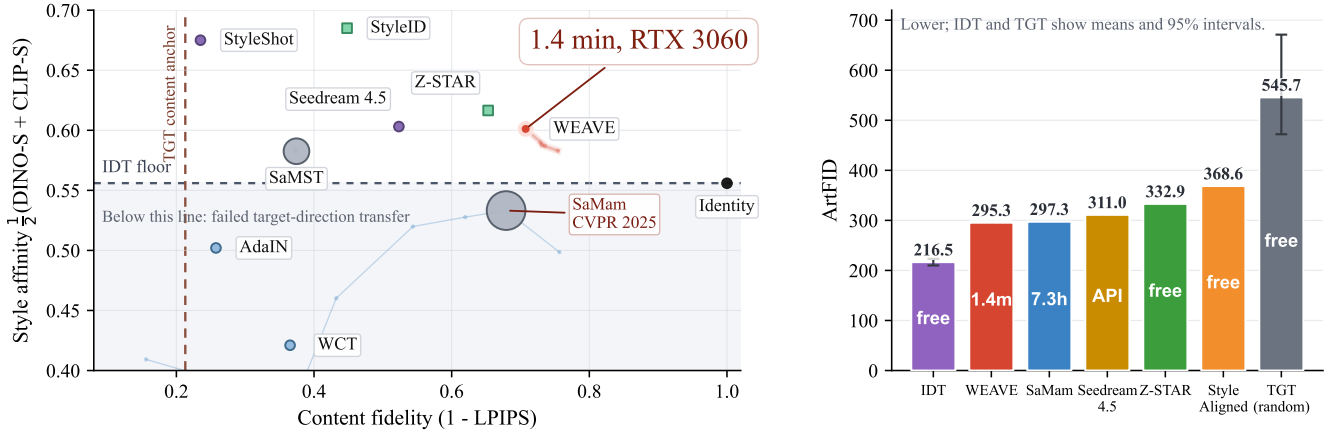


Figure 1: IDT–TGT sandwich comparison on Distinct5-WikiArt. Left: the two axes represent style affinity and content preservation; bubble area denotes training time and red marks WEAVE. IDT and TGT provide the displayed style and content references. Right: target-pooled ArtFID on the same benchmark. IDT shows the board estimate and a 95% target-style bootstrap interval; TGT shows the mean and 2.5–97.5 percentile interval over 30 reference draws. Bar labels give training time.

Introduction

Style transfer faces two linked problems in deployment and evaluation. Recent image foundation models such as Qwen-Image and FLUX.2 provide strong generation and editing priors (Wu et al. 2025; Black Forest Labs 2025), but can require more than 60 GB of GPU memory. Lightweight alternatives avoid this cost, but some preserve the source so strongly that their style affinity falls below the unchanged

input. Conventional protocols can hide this no-op behavior: LPIPS rewards source reconstruction, and the content factor of ArtFID can favor a near-copy, yet the identity mapping is rarely reported as an explicit baseline.

Figure 1 makes the missing comparison visible with two simple outputs. Identity (IDT) returns the source unchanged, while Target Style (TGT) returns an unrelated reference from the requested style. A useful transfer should move beyond

IDT in the style direction without moving as far from the source as TGT in the content direction. The two references expose opposite failures in the original metric units: methods below IDT have not acquired enough target style, whereas methods beyond TGT have discarded at least as much source content as a source-independent style image. We call this the IDT–TGT sandwich. We report this diagnostic alongside DINO, CLIP, LPIPS, and visual comparisons.

Why do compact models enter the identity regime? Our probes identify the shared coordinate system as one contributor, separate from parameter count. Neural networks learn low frequencies first (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place much more energy in global layout and object shape than in brushstroke and fine texture. A velocity model trained on the full latent must explain both with one field. The resulting low-frequency residuals are larger and easier to fit, so their gradients dominate the update and weaken style-conditioned high-frequency motion. Frequency-domain translation reports a related content–style conflict (Cai et al. 2021); our gradient and subband probes measure it directly in the trained model. Simply increasing width does not change this competition because both signals still share the same basis and objective.

We therefore propose WEAVE, which changes the representation and target before increasing capacity. A frozen diffusion VAE supplies a compact latent, while a one-level Haar transform separates low-frequency structure (LL) from oriented texture (LH/HL/HH). A 1.04M-parameter velocity model learns source-aligned transport with weaker LL supervision; compact orientation-specific target codes modulate the matching LH/HL heads; and one stepwise AdaIN update supplies target-style statistics to the high-frequency residual. Cached latents remove the VAE encoder and diffusion U-Net from training; eight compact Euler evaluations and one VAE decode are sufficient at inference. An internal gradient transition selects the fourth epoch without image evaluation, giving a 1.4-minute run on one RTX 3060; the same Haar statistical operator can be added to a frozen SD1.5 editor without retraining.

Contributions. (1) **Evaluation:** We advocate the IDT–TGT sandwich, which reports a no-op reference and a source-independent target-style reference alongside standard metrics, exposing both style suppression and content collapse. (2) **Representation and method:** We measure low-frequency gradient dominance in a shared latent objective and develop a wavelet-decoupled transport that assigns structure, learned texture transport, and trajectory-level statistics to explicit paths. (3) **Efficiency and evidence:** WEAVE uses 1.04M trainable parameters and 1.4 minutes of single-GPU training. Cross-dataset evaluation, internal gradient probes, and a frozen SD1.5 plugin provide empirical checks beyond the main board.

Related Work

Neural Style Transfer

Artistic style transfer was introduced by optimizing pixels to match content features and Gram statistics of a pretrained network (Gatys, Ecker, and Bethge 2015, 2016). AdaIN aligns

channel-wise moments (Huang and Belongie 2017), while WCT aligns a full feature covariance (Li et al. 2017); both replace iterative optimization with a feed-forward feature transform. SANet and transformer-based variants add correspondence or long-range attention (Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Kwon et al. 2024), and CUT learns unpaired translation through patch-level contrast (Park et al. 2020). Training-free methods also reuse pretrained codebooks or hierarchical vision transformers (Gim, Yoo, and Uh 2024; Zhang et al. 2024a). The survey of Zhang et al. (Zhang and Tang 2025) summarizes this progression. These methods differ in supervision and backbone, but most still mix structure and appearance in one feature tensor. Stronger moment or attention transfer can therefore alter source geometry; WEAVE instead restricts the analytic style correction to explicit frequency coordinates.

Diffusion-based and Training-free Style Editing

Latent diffusion models (Rombach et al. 2022) move editing into a compressed generative space and provide strong semantic priors. Prompt-to-Prompt and SDEdit control generation through attention or noising (Hertz et al. 2022; Meng et al. 2021); StyleID and StyleAligned modify attention statistics (Chung, Hyun, and Heo 2024; Hertz et al. 2024); IP-Adapter adds an image prompt (Ye et al. 2023); and InST, DiffuseIT, ArtBank, and Z-STAR use inversion or pretrained features for style transfer (Zhang et al. 2023; Kim, Kwon, and Ye 2023; Zhang et al. 2024b; Lu et al. 2024). These methods are often called training-free because they do not update backbone weights, but inference still invokes a large denoiser repeatedly and may require inversion. Recent Qwen-Image and FLUX.2 illustrate the resulting deployment gap between large image backbones and compact transfer models (Wu et al. 2025; Black Forest Labs 2025). WEAVE retains only the frozen VAE and learns the remaining transport with a compact model; the SD1.5 plugin separately tests whether the analytic Haar operator transfers back to a large pretrained editor.

Flow Matching and Schrödinger Bridges

Rectified flow (Liu, Gong, and Liu 2023) and flow matching (Lipman et al. 2022) regress a velocity along an interpolation between endpoint distributions, enabling short numerical paths. Schrödinger bridges (Léonard 2014) add entropy-regularized stochastic transport, and DSB, DSBM, I2SB, and DDIB develop practical diffusion or flow solvers for unpaired domains (De Bortoli et al. 2021; Shi et al. 2023; Liu et al. 2023; Su et al. 2023). These methods are related to regularized optimal transport (Cuturi 2013; Peyré and Cuturi 2019), but their image objectives usually measure all coordinates together. WEAVE uses endpoint-interpolation velocity regression with a source-aligned endpoint, an orthogonal wavelet basis, and separately weighted blocks. This retains a short numerical trajectory while changing which errors compete during learning.

Lightweight Models and Evaluation

Mamba-based SaMST and SaMam reduce trainable size and sequence cost (Liu et al. 2024, 2025; Gu and Dao 2023),

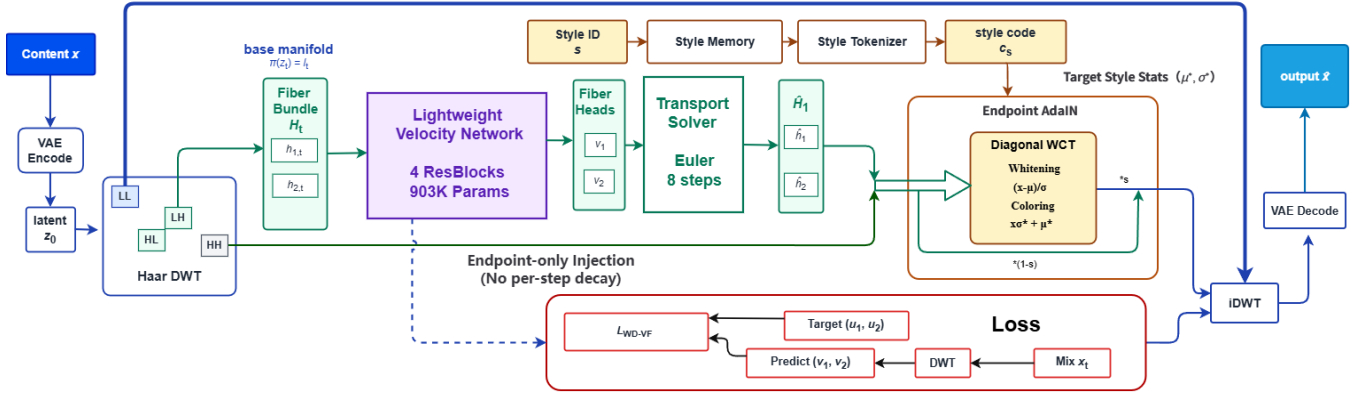


Figure 2: **WEAVE architecture.** A single-level Haar transform exposes LL and three high-frequency bands. The compact velocity network predicts LL/LH/HL motion, while pooled target LH/HL codes modulate their matching residual heads; HH has no learned velocity head. After every Euler update, high-frequency AdaIN aligns the combined high-frequency residual to the target-style statistics. The complete D5 model has 1.04M trainable parameters.

showing that style transfer need not inherit a full diffusion U-Net. Efficiency alone, however, does not establish that the output moved toward the requested style. LPIPS measures perceptual source distance (Zhang et al. 2018), and ArtFID combines style distribution and content terms (Wright and Ommer 2022); both can reward a near-copy when the identity output is omitted. IDT makes this no-op score explicit. TGT adds the opposite reference: a target-style image with no source dependence. Reporting both rows preserves the standard metrics and their raw units, while identifying whether an apparent gain comes from style suppression or content collapse. The references are benchmark-specific diagnostics rather than universal thresholds.

Spectral Bias and Frequency-aware Optimization

Neural networks preferentially fit low-frequency functions (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place most spectral energy in coarse structure. The two effects are distinct but reinforce each other: the training target supplies larger low-frequency residuals, while the network fits those residuals earlier. Frequency-domain translation therefore constrains low and high frequencies separately to protect content identity (Cai et al. 2021). FreqFlow instead introduces time-dependent frequency weighting and a dedicated frequency branch for class-conditional flow-based image generation (Ren et al. 2026). Our contribution is to examine this optimization imbalance in a compact velocity model through per-band loss, input gradients, head gradients, and endpoint interventions. The measurements motivate a structural preconditioner for source-conditioned, unpaired style transfer rather than unconditional generation.

Wavelet and Latent-space Alignment

Wavelets separate local information by scale and orientation and support exact reconstruction. They have been used in restoration and generation (Phung, Dao, and Tran 2023) and in photorealistic style transfer (Yoo et al. 2019). The Haar basis is particularly simple: its orthogonality gives additive

energy decomposition, and its compact support preserves local correspondence (Daubechies 1992). VAE latents provide a complementary reduction in spatial cost (Rombach et al. 2022), while AdaIN and WCT provide closed-form statistical alignment. WEAVE combines these ingredients inside the transport objective rather than using wavelets only as preprocessing or postprocessing. LL, LH, and HL receive explicit velocity roles; HH is handled only by stepwise statistical alignment. This division is the central difference from a single latent-space transform such as Latent-WCT.

Method

A direct content-to-style target changes layout, palette, and texture at once. Under shared latent MSE, the largest residual can dominate the update even when it is not the most useful signal for style transfer. WEAVE instead constructs a low-frequency-aligned target, learns band-weighted transport, and corrects high-frequency statistics along the inference path (Figure 2).

Why Split the Transport

In a shared latent objective, coarse structural residuals and style residuals compete for the same capacity. Our probes show larger LL transport energy and shared-trunk gradients, while style separation is stronger in high-frequency bands (Figure 3). WEAVE therefore changes the LL target, assigns separate output heads, and downweights—but does not remove—the LL regression term. The resulting design follows the measured optimization imbalance in this model.

Framework Setup

Let $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times H \times W}$ and $z_s = \mathcal{E}(x_s)$ be content and target-style latents from the frozen SD1.5 VAE (Rombach et al. 2022). A direct path from z_c to z_s also learns the unrelated geometry of x_s . We instead decompose $\mathcal{W}(z_c) = (\ell_c, h_{1,c}, h_{2,c}, h_{3,c})$ and similarly for z_s , then form the low-

frequency-aligned endpoint

$$\begin{aligned}\tilde{\ell}_s &= (1 - \alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), \quad \alpha = 0.3, \\ z_1^* &= \mathcal{W}^{-1}(\tilde{\ell}_s, h_{1,s}, h_{2,s}, h_{3,s}).\end{aligned}\quad (1)$$

The LL blend scales the supervised coarse movement while retaining the source coordinate as its anchor; the high-frequency target comes from the requested style. Exact source LL suppresses palette change, whereas the complete style latent would also supervise unrelated target geometry.

Thus α scales the supervised coarse movement, while the target high-frequency bands retain an appearance signal. This target construction is distinct from reducing the LL loss weight: the former changes the requested velocity and the latter changes its optimization pressure.

Training samples a lightly perturbed endpoint interpolation

$$z_t = (1 - t)z_c + tz_1^* - \sigma\sqrt{t(1 - t)}\epsilon, \quad u = z_1^* - z_c, \quad (2)$$

where $t \sim \mathcal{U}([0, 1])$, $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma = 0.02$. The perturbation regularizes the input state and vanishes at both endpoints; the supervision remains the clean endpoint displacement u . At inference, the perturbation is absent. Because $\mathcal{W}$ is linear, the interpolation decomposes into the same sub-bands. WEAVE changes the endpoint construction, parameterization, and frequency weights of this velocity regression.

Wavelet Decomposition and Frequency Routing

Haar wavelet decomposition. A one-level Haar transform gives $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$: one LL band and three high-frequency bands. Haar is orthonormal, local, and exactly invertible, which lets us assign losses and prediction heads to explicit coordinates (Daubechies 1992). This separation is exact in latent space, although the nonlinear VAE decoder can still couple visible color, texture, and structure. For each 2×2 latent block (a, b, c, d) , our transform is

$$\begin{bmatrix} \ell \\ h_1 \\ h_2 \\ h_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}. \quad (3)$$

The rows are orthonormal, so the transform preserves energy and the inverse is its transpose. LL records the block average; LH, HL, and HH record vertical, horizontal, and diagonal changes, respectively. This is the precise sense in which WEAVE “separates frequencies”; it does not assume that each band contains only one semantic factor.

Affine-fiber view. For $\mathcal{W}(z) = (\ell_z, h_{1,z}, h_{2,z}, h_{3,z})$, define $P_L z = \mathcal{W}^{-1}(\ell_z, 0, 0, 0)$ and $P_H = I - P_L$. Let $\mathcal{Z}_L$ be the range of P_L and $\mathcal{Z}_H = \ker P_L$ its orthogonal complement. Haar gives the direct sum $\mathcal{Z} = \mathcal{Z}_L \oplus \mathcal{Z}_H$. For a fixed low-frequency coordinate ℓ , the corresponding affine fiber is

$$\mathcal{F}_\ell = \{z : P_L z = \ell\} = \mathcal{W}^{-1}(\ell, 0, 0, 0) + \mathcal{Z}_H. \quad (4)$$

Latents in $\mathcal{F}_\ell$ share the same Haar LL coordinate but may differ in local appearance after decoding. WEAVE does not keep ℓ exactly fixed: the aligned target and the $\lambda_{LL} = 0.3$ loss allow a limited coarse-color motion. In contrast, the stepwise AdaIN operator acts on $P_H z$ and then reattaches the current low-pass component. We use “fiber” for this exact affine decomposition.

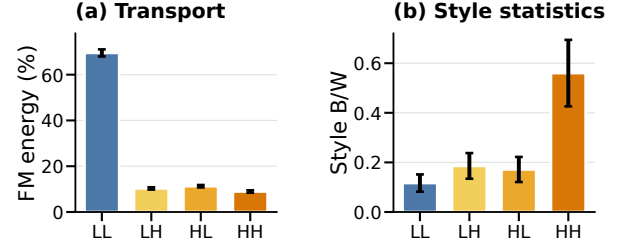


Figure 3: **Frequency probes.** (a) Share of cross-style endpoint-displacement energy in each Haar band. (b) Between-style/within-style variance ratio of per-image latent statistics. Error bars show the standard error across source–target style routes in (a) and across style families in (b).

Empirical frequency probes. Across 5,000 cross-style latent pairs, transport energy concentrates in LL while style separation is strongest in HH (Figure 3). This motivates a weak LL loss and greater emphasis on high-frequency prediction.

Architecture. The velocity network uses four width-64 residual blocks and separate 1×1 heads for LL, LH, and HL. Pooled LH/HL codes modulate only their matching heads, preserving target appearance without passing target coordinates into the LL path. WEAVE does not learn an HH velocity head; HH is handled by the statistical update below.

Why it trains quickly. Each pair provides a direct endpoint displacement at every sampled time, while stepwise AdaIN supplies high-frequency moments analytically. Training therefore regresses a small latent-space velocity field rather than unrolling a denoiser, inversion, or adversarial game. The frozen VAE, cached latents, and eight-step solver account for the measured cost.

Spectral velocity-matching objective. Let $(v_\ell, v_{h_1}, v_{h_2}) = v_\theta(z_t, t, s)$ denote the bandwise velocity predicted from the current latent, time, and target-style condition, and let $(u_\ell, u_{h_1}, u_{h_2}, u_{h_3}) = \mathcal{W}(u)$. The spectral velocity-matching term is

$$\begin{aligned}\mathcal{L}_{\text{SFM}} &:= \mathbb{E}_t [\lambda_{LL} \|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 \\ &\quad + \|v_{h_2} - u_{h_2}\|_2^2],\end{aligned}\quad (5)$$

with $\lambda_{LL} = 0.3$. The lower LL weight limits structural dominance while still allowing coarse tone to move. The distinct output heads remove output-level sharing, while the common trunk can still couple the bands. HH is omitted because WEAVE has no HH velocity head. The ablation in Table 2 shows similar performance with $\lambda_{LL} = 1.0$.

Stepwise High-frequency Statistical Alignment

Velocity matching organizes LL/LH/HL motion but leaves a weak appearance signal. WEAVE therefore applies AdaIN to the combined high-frequency residual after every Euler update. Using the projections defined above, for a current residual $q = P_H(z)$ and target residual $q_s = P_H(z_s)$, define (Huang

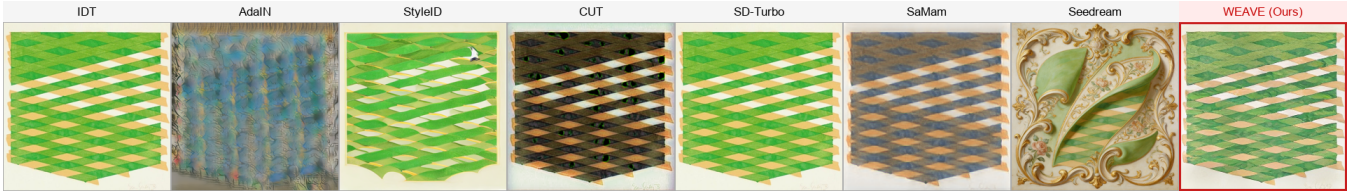


Figure 4: Extreme cross-domain stylization (Minimalism $\rightarrow$ Rococo). Diffusion editors substantially alter source structure, while several lightweight or classical outputs remain close to identity. WEAVE retains the source topology while adding the target’s painterly appearance.

and Belongie 2017)

$$\mathcal{A}(q; q_s) = \frac{q - \mu(q)}{\sigma(q)} \sigma(q_s) + \mu(q_s), \quad (6)$$

where μ and σ are channel-wise spatial statistics. One inference step is

$$\begin{aligned} \bar{z}^{k+1} &= z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h_1}^k, v_{h_2}^k, 0), \\ z^{k+1} &= P_L(\bar{z}^{k+1}) + (1 - \beta)q^{k+1} + \beta \mathcal{A}(q^{k+1}; (1 + \eta)q_s), \end{aligned} \quad (7)$$

with $q^{k+1} = P_H(\bar{z}^{k+1})$, $K = 8$, $\beta = 2.0$, and $\eta = 0.1$ in our experiments. Since $\beta > 1$, the update extrapolates from the current residual toward the matched target statistics. Applying it at every step makes statistical alignment part of the inference dynamics rather than a post-processing operation. The learned LH/HL path provides organized motion, while the same update supplies statistics to HH. The matched control confirms that removing AdaIN moves the result toward source preservation.

Training and Inference

Training minimizes Eq. 5. Inference uses eight Euler steps and the frozen VAE decoder. Optimization, caching, probe, and evaluation settings are given in the supplement.

Experiments

Setup and Protocol

We evaluate D5 (750 requests), lower-resolution P2A, and the 20-family R5 benchmark. DINO-S is the primary style metric; CLIP-S, DINO-C, and LPIPS report complementary style and content views. IDT and TGT provide the no-op and target-copy anchors; full benchmark construction, preprocessing, and fixed evaluation settings are in the supplement.

Main Results

Reading the sandwich. The marks expose methods that remain below IDT on style or cross TGT on content distance. The bracket is benchmark- and metric-specific: DINO-S/CLIP-S capture different style cues, while LPIPS/DINO-C capture different content failures. We therefore report each raw metric and compute IDT/TGT for each benchmark rather than collapsing them into one score.

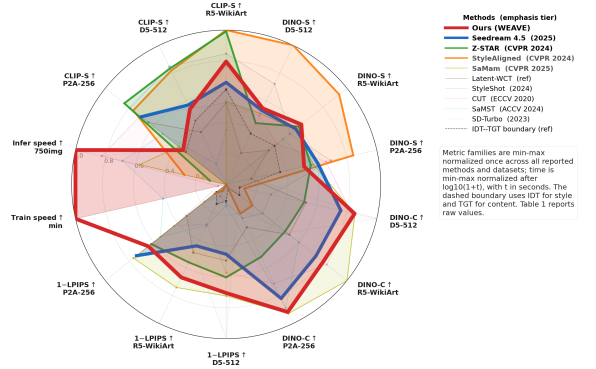


Figure 5: Style, content, and cost across three benchmarks. The dashed boundary uses IDT on style axes and TGT on content axes. Each metric family is min–max normalized once across all reported methods and datasets, so its best reported value reaches the outer ring; speed uses the same normalization after $\log_{10}(1 + t)$, where t is seconds. Table 1 reports raw values and checkpoint provenance.

D5 result. WEAVE clears IDT on both style measures while retaining substantially more source content than editors with a similar or stronger style score. Among trained methods that clear the CLIP-S identity reference, it has the lowest LPIPS and uses only 1.04M trainable parameters. The selected high-frequency conditioning improves both DINO-S and CLIP-S over the base configuration without crossing the TGT content reference. The comparison with Seedream is especially informative: WEAVE reaches higher DINO-S with substantially lower content cost.

P2A and R5. WEAVE remains inside the bracket on P2A. On R5, it improves DINO-S over IDT, CUT, and SaMam while retaining substantially more source content than StyTR-2 and AesPA-Net. The stronger D5-512 profile is consistent with native high-frequency coefficients preserving detail compressed at 256. Because D5 and P2A also differ in data, we treat this as scale potential rather than an isolated resolution claim. Figure 5 summarizes the reported operating points.

Cost. WEAVE is substantially cheaper to train and run than the learned baselines in Table 1. Latent-WCT is faster still but falls below IDT, showing that analytic wavelet statistics alone do not replace learned transport.

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	M	min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
Target Style	1.000	0.863	0.776	0.215	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s
SD-Turbo	0.484	0.693 $\dagger$	0.003	0.922	0.341 $\dagger$	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 ⁺¹¹⁰⁰	free	5 m
StyleAligned	0.675	0.780	0.869 $\ddagger$	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 $\ddagger$	0.315	0 ⁺¹¹⁰⁰	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	0.822	0.384	0.526	0 ⁺¹¹⁰⁰	free	3 h
StyleShot	0.563	0.787	0.765 $\ddagger$	0.377	0.552	0.774	0.713	0.335	0.484	0.787	0.795 $\ddagger$	0.380	0 ⁺²⁴⁰⁰	free	5.1 h
StyleID	0.548	0.822	0.552	0.416	0.256 $\ddagger$	0.648 $\ddagger$	0.691	0.147 $\ddagger$	0.571	0.815	0.493	0.374	0 ⁺¹¹⁰⁰	free	63 m
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 $\ddagger$	0.620	0.795	7.0	322.6	5 m
SaMST	0.271 $\dagger$	0.618 $\dagger$	0.749 $\ddagger$	0.145 $\ddagger$	0.501	0.709	0.398	0.838	0.461	0.667 $\ddagger$	0.612	0.615	8.3	39.5	10 m
SaMam	0.477	0.582 $\dagger$	0.243	0.812	0.505	0.677	0.205	0.870	0.503	0.712 $\dagger$	0.227	0.925	8.8	436.0	17.6 m
StyTR-2	0.501	0.709	0.596	0.747	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 ^{+12.9}	$\sim$ 1440*	38 m
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 ^{+20.0}	–	23 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362 $\dagger$	0.673 $\dagger$	0.441	0.559	0.338 $\dagger$	0.738	0.585	0.386	0.414 $\dagger$	0.710 $\dagger$	0.444	0.553	0	free	18 s
WEAVE	0.4918	0.7128	0.2595	0.8102	0.4801	0.6681	0.3116	0.8612	0.5226	0.7747	0.2895	0.7717	1.04 ⁺⁸⁴	1.4	126 s

Table 1: Main comparison. Higher DINO-S, CLIP-S, and DINO-C and lower LPIPS are better. Excluding IDT and TGT, the best two entries in each column are bold; first place has a double underline and second place a single underline. Params and Train rank trained methods, while Infer ranks reported runtimes. $\dagger$: style does not exceed IDT; $\ddagger$: content falls outside TGT. WEAVE uses its selected D5 checkpoint for D5 and dataset-specific base checkpoints for P2A/R5; its cost columns refer to the selected D5 configuration. Parameters are in millions, with frozen parameters shown as superscripts. Unless marked otherwise, times are measured on one RTX 3060; * is the StyTR-2 author report (about one day on two P100 and two RTX 3090 GPUs (Deng et al. 2022)).

Variant	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
WEAVE	0.4918	0.7128	0.2595	0.8102
w/o AdaIN	0.4821	0.7005	0.2258	0.8455
w/o HF conditioning	0.4825	0.7157	0.2837	0.8125
$\lambda_{LL}=1.0$	0.4910	0.7180	0.2626	0.7874
direct endpoint	0.4894	0.7172	0.3138	0.7966
learned HH head	0.4930	0.7164	0.2670	0.8061

Table 2: Component ablation on the same 750 D5 requests. The no-AdaIN control uses the selected epoch-4 checkpoint; retrained variants use the same fixed-latent selection rule.

Mechanistic Ablations

Table 2 tests the endpoint, frequency weighting, high-frequency conditioning, stepwise AdaIN, and HH prediction. Removing AdaIN lowers DINO-S by 0.0097 and CLIP-S by 0.0123 while moving the output closer to the source. Removing high-frequency conditioning similarly lowers DINO-S and increases LPIPS. Replacing the source-aligned endpoint with the direct target has the largest content cost, increasing LPIPS from 0.2595 to 0.3138. In contrast, changing the nonzero LL weight from 0.3 to 1.0 changes peak DINO-S by less than 0.001, showing that the method does not depend on a narrow weight choice. A learned HH head gives a small style gain with a small content cost, so we retain the simpler model. Full training curves and configuration checks are in the supplement.

Robustness Checks

Representation and adaptation. Supplementary controls compare latent and RGB transport and evaluate adaptation to

held-out style families. They support the latent representation and show that updating the compact style memory captures most of the gain from full fine-tuning.

The relative fixed-latent criterion selects epochs 4, 4, and 3 for seeds 42, 7, and 123, respectively, close to the observed DINO-S peaks. Matched LL-weight and HH variants show the same transition. Complete trajectories and the fixed rule are provided in the supplement.

Reference-pool stability. We resample one without-replacement reference pool per target style and reuse it for all requests to that style. The board-mean DINO-S margin over IDT remains positive across the tested pool sizes and draws, showing that the aggregate conclusion is not determined by a particular reference subset. This test does not assert that every individual request improves; pair-level statistics are provided in the supplement.

Unseen families. On five unseen WikiArt families, WEAVE gives the highest CLIP-S and DINO-C and the lowest LPIPS (Table 3); SaMam is 0.015 higher on DINO-S but has higher LPIPS.

Method	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-S $\uparrow$	DINO-C $\uparrow$
WEAVE	0.7742	0.2802	0.5313	0.7587
SaMam	0.7577	0.3990	0.5466	0.7369
SaMST	0.7485	0.6213	0.3913	0.2680

Table 3: Zero-shot transfer to five held-out WikiArt style families (Other5; 750 pairs) using the D5 checkpoint.

SD1.5 editor	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
Base	0.3974	0.7221	0.3591	0.6589
+ wavelet alignment	0.4080	0.7366	0.3630	0.6531

Table 4: Plug-in wavelet alignment for a frozen SD1.5 editor on 600 cross-style D5 pairs. No SD1.5 weights are updated.

Basis and resolution. An alternative wavelet gives a similar style–content trade-off, so we retain Haar for its orthogonality, compact support, and lower implementation cost. Training at the native resolution performs better than upsampling a lower-resolution checkpoint, further supporting the scale potential observed in the main comparison. Detailed controls are in the supplement.

Plug-in Extension for a Frozen Diffusion Editor

We apply the same Haar-domain statistical alignment to a frozen SD1.5 image-to-image pipeline without updating its weights. The base editor uses an empty prompt, strength 0.5, and 20 DDIM steps; the plugin aligns LH/HL/HH at scale 0.8 while leaving LL unchanged. On 600 cross-style D5 pairs, CLIP-S rises from 0.7221 to 0.7366 and DINO-S from 0.3974 to 0.4080 (both $p < 10^{-15}$, paired t -test), while LPIPS changes from 0.3591 to 0.3630 (Table 4). The gains remain on all 750 pairs. The DWT–AdaIN–IDWT operation adds 4.9 ms per image and requires no model update.

Discussion

Subband and gradient measurements locate the optimization imbalance in WEAVE, while matched interventions separate the effects of source alignment, high-frequency conditioning, and statistical alignment. Stability across nonzero LL weights further indicates that the gain comes from the frequency-specific construction rather than a sharply tuned scalar.

Limitations

WEAVE is evaluated with one SD1.5 VAE and a single Haar level. Styles dominated by global palette or layout changes, and styles expressed across several scales, may require a deeper decomposition or a different tokenizer. Haar orthogonality separates latent coordinates, not semantic factors: LH/HL can still contain geometric edges, and the nonlinear decoder can couple visible structure and appearance. The compact model is trained for a style collection rather than used as a universal pretrained editor. Its learned velocity omits HH; the matched ablation shows a small style–content trade-off, and we retain the simpler route.

The fixed-latent criterion tracks the style peak in the selected run, two additional seeds, and several matched controls, but this evidence is limited to one architecture and three seeds; it is not a general convergence test. IDT and TGT are dataset-specific references, while DINO, CLIP, LPIPS, and ArtFID remain imperfect perceptual proxies. In particular, the content term in ArtFID can favor outputs near the source, so we use it as a complementary plausibility measure rather than a style ranking. Baselines use their released or recommended operating points rather than a joint search under one compute

budget, so the table compares practical configurations rather than compute-matched optima. Finally, the SD1.5 extension demonstrates transfer of the wavelet operator to one frozen editor; broader diffusion backbones have not been tested.

Conclusion

The IDT–TGT sandwich gives style and content metrics an explicit direction: IDT identifies methods stuck near the source, while TGT identifies those that abandon it. WEAVE addresses this conflict in frequency coordinates. A source-aligned target limits unnecessary structure motion, band-weighted flow learns organized transport, and stepwise AdaIN supplies high-frequency statistics without repeatedly perturbing the trajectory.

On the reported benchmark configurations, WEAVE exceeds the identity style reference while retaining content closer to the source than TGT. Gradient and subband measurements are consistent with low-frequency dominance in this model, and matched interventions distinguish the effects of source alignment, high-frequency conditioning, and stepwise statistics. The higher-resolution setting and the frozen-SD1.5 experiment suggest possible scale and transfer directions. The same appearance–geometry tension also arises in synthetic-to-real translation (Zang, Hu, and Song 2026), suggesting structure-preserving latent augmentation as future work.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., “Impressionism”) rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not model or label individual artists, although inference uses target-style reference images. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. The D5 run stops after four epochs and takes 82.8 seconds (1.4 minutes) on a single RTX 3060 (12 GB; batch 96), without online image evaluation. Generation-only inference for 750 pairs takes 126 seconds for the selected WEAVE model (8-step ODE + VAE decode), and a full metrics pass adds metric extraction and writing overhead. The reported training time starts after latent caching. The model has 1.04M trainable parameters plus the frozen 84M VAE. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

References

Black Forest Labs. 2025. FLUX.2: Frontier Visual Intelligence. <https://bfl.ai/blog/flux-2>.

- Cai, M.; Zhang, H.; Huang, H.; Geng, Q.; Li, Y.; and Huang, G. 2021. Frequency Domain Image Translation: More Photo-Realistic, Better Identity-Preserving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Cuturi, M. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. *Advances in Neural Information Processing Systems*.
- Daubechies, I. 1992. Ten Lectures on Wavelets. *SIAM*.
- De Bortoli, V.; Thornton, J.; Heng, J.; and Doucet, A. 2021. Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling. *arXiv preprint arXiv:2106.01357*.
- Deng, Y.; Tang, F.; Dong, W.; Sun, W.; Huang, F.; and Xu, C. 2022. StyTr2: Image Style Transfer with Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2015. A Neural Algorithm of Artistic Style. In *arXiv preprint arXiv:1508.06576*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Gim, H.; Yoo, J.; and Uh, Y. 2024. Content-Adaptive Style Transfer: A Training-Free Approach with VQ Autoencoders. In *Proceedings of the Asian Conference on Computer Vision*.
- Gu, A.; and Dao, T. 2023. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv preprint arXiv:2312.00752*.
- Hertz, A.; Mokady, R.; Tenenbaum, J.; Aberman, K.; Pritch, Y.; and Cohen-Or, D. 2022. Prompt-to-Prompt Image Editing with Cross Attention Control. *arXiv preprint arXiv:2208.01626*.
- Hertz, A.; Voynov, A.; Fruchter, S.; and Cohen-Or, D. 2024. Style Aligned Image Generation via Shared Attention. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Hong, K.; Jeon, S.; Yang, H.; Fu, J.; and Byun, H. 2023. AesPA-Net: Aesthetic Pattern-Aware Style Transfer Networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. In *ICCV*.
- Kim, G.; Kwon, T.; and Ye, J. C. 2023. Diffusion-based Image Translation using Disentangled Style and Content Representation. In *International Conference on Learning Representations*.
- Kwon, J.; Kim, S.; Lin, Y.; Yoo, S.; and Cha, J. 2024. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Léonard, C. 2014. A Survey of the Schrödinger Problem and Some of Its Connections with Optimal Transport. *Discrete and Continuous Dynamical Systems - A*.
- Li, Y.; Fang, C.; Yang, J.; Wang, Z.; Lu, X.; and Yang, M.-H. 2017. Universal Style Transfer via Feature Transforms. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2022. Flow Matching for Generative Modeling. *arXiv preprint arXiv:2210.02747*.
- Liu, G.-H.; Vahdat, A.; Huang, D.-A.; Theodorou, E. A.; Nie, W.; and Anandkumar, A. 2023. I2SB: Image-to-Image Schrödinger Bridge. In *International Conference on Machine Learning*.
- Liu, H.; Wang, L.; Guan, W.; Zhang, Y.; and Guo, Y. 2024. Pluggable Style Representation Learning for Multi-Style Transfer. In *Proceedings of the Asian Conference on Computer Vision*.
- Liu, H.; Wang, L.; Zhang, Y.; Yu, Z.; and Guo, Y. 2025. SaMam: Style-aware State Space Model for Arbitrary Image Style Transfer. *arXiv preprint arXiv:2503.15934*.
- Liu, X.; Gong, C.; and Liu, Q. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *International Conference on Learning Representations*.
- Lu, Z.-S.; Vukoti, S.; Hadji, I.; Dorken, W.; and Levi, G. 2024. Z-STAR: Training-Free Zero-Shot Style Transfer via Attention Rearrangement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Meng, C.; He, Y.; Song, Y.; Song, J.; Blackwell, J.; Jia, J.; and Ermon, S. 2021. SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations. In *Proceedings of the IEEE International Conference on Computer Vision*.
- Park, D. Y.; and Lee, K. H. 2019. Arbitrary Style Transfer with Style-Attentional Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Park, T.; Efros, A. A.; Zhang, R.; and Zhu, J.-Y. 2020. Contrastive Learning for Unpaired Image-to-Image Translation. In *Proceedings of the European Conference on Computer Vision*.
- Peyré, G.; and Cuturi, M. 2019. Computational Optimal Transport. *Foundations and Trends in Machine Learning*.
- Phung, H.; Dao, Q.; and Tran, A. 2023. Wavelet Diffusion Models Are Fast and Scalable Image Generators. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Rahaman, N.; Baratin, A.; Arpit, D.; Draxler, F.; Li, M.; Hamprecht, F.; Bengio, Y.; and Courville, A. 2019. On the Spectral Bias of Neural Networks. In *ICML*.
- Ren, S.; Yu, Q.; He, J.; Shen, X.; Yuille, A.; and Chen, L.-C. 2026. Frequency-Aware Flow Matching for High-Quality Image Generation. *arXiv:2604.15521*.
- Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *CVPR*.

Shi, Y.; De Bortoli, V.; Campbell, A.; and Doucet, A. 2023. Diffusion Schrödinger Bridge Matching. In *Advances in Neural Information Processing Systems*.

Su, X.; Song, J.; Meng, C.; and Ermon, S. 2023. Dual Diffusion Implicit Bridges for Image-to-Image Translation. In *International Conference on Learning Representations*.

Wright, M.; and Ommer, B. 2022. ArtFID: Quantitative Evaluation of Neural Style Transfer. *arXiv preprint arXiv:2207.12280*.

Wu, C.; Li, J.; Zhou, J.; Lin, J.; Gao, K.; Yan, K.; ming Yin, S.; Bai, S.; Xu, X.; Chen, Y.; Chen, Y.; Tang, Z.; Zhang, Z.; Wang, Z.; Yang, A.; Yu, B.; Cheng, C.; Liu, D.; Li, D.; Zhang, H.; Meng, H.; Wei, H.; Ni, J.; Chen, K.; Cao, K.; Peng, L.; Qu, L.; Wu, M.; Wang, P.; Yu, S.; Wen, T.; Feng, W.; Xu, X.; Wang, Y.; Zhang, Y.; Zhu, Y.; Wu, Y.; Cai, Y.; and Liu, Z. 2025. Qwen-Image Technical Report. *arXiv:2508.02324*.

Xu, Z.-Q. J.; Zhang, Y.; and Luo, T. 2022. Overview frequency principle/spectral bias in deep learning. *arXiv preprint arXiv:2201.07395*.

Ye, H.; Zhang, J.; Liu, S.; Han, X.; and Yang, W. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. *arXiv preprint arXiv:2308.06721*.

Yoo, J.; Uh, Y.; Chun, S.; Kang, B.; and Ha, J.-W. 2019. Photorealistic Style Transfer via Wavelet Transforms. In *ICCV*.

Zang, C.; Hu, J.; and Song, X. 2026. FD-DB: Frequency-Decoupled Dual-Branch Network for Unpaired Synthetic-to-Real Domain Translation.

Zhang, C.; Xu, X.; Wang, L.; Dai, Z.; and Yang, J. 2024a. S2WAT: Image Style Transfer via Hierarchical Vision Transformer Using Strips Window Attention. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *CVPR*.

Zhang, T.; and Tang, H. 2025. Style Transfer: A Decade Survey. *arXiv preprint arXiv:2506.19278*.

Zhang, Y.; Huang, N.; Tang, F.; Huang, H.; Ma, C.; Dong, W.; and Xu, C. 2023. Inversion-Based Style Transfer with Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhang, Y.; et al. 2024b. ArtBank: Artistic Style Transfer with Pre-trained Diffusion Model and Implicit Style Prompt. In *Proceedings of the AAAI Conference on Artificial Intelligence*.